

Tu(ğ)ba Hato

PHD CANDIDATE · HEIDELBERG UNIVERSITY

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Research Interests

Quantitative psychologist focused on Bayesian cognitive modeling, simulation studies for model evaluation, and the analysis of individual and intra-individual processes in large-scale behavioral data.

Education

Heidelberg University

PHD PSYCHOLOGICAL METHODS

- Advisor: Dr. Andreas Voss

Heidelberg

2022 – present

Middle East Technical University (ODTÜ|METU)

M.SC COGNITIVE PSYCHOLOGY

- Advisor: Dr. Aslı Kılıç
- Thesis title: Testing Encoding and Retrieval Dynamics of Pairs Using Probed Recall Task

Ankara

2020 – 2022

Middle East Technical University (ODTÜ|METU)

B.SC PSYCHOLOGY

Ankara

2016 – 2020

Professional Experience

2022- **PhD Candidate**, Psychological Methods, Heidelberg University

2022 **TUBITAK Project Assistant**, Middle East Technical University

2021-2022 **Cognitive Psychologist**, Neo Auvra

2019 **Undergraduate Research Intern**, Cognitive Sciences, Middle East Technical University

2018-2020 **Undergraduate Research Assistant**, Department of Psychology, Middle East Technical University

Scientific Programming & Skills

Python: Data manipulation with *numpy* and *pandas*, visualization with *matplotlib*. Modeling with SBI approaches via *BayesFlow*, machine learning in *skicit-learn*.

R: Data wrangling with *tidyverse*, visualization with *ggplot2*, and data analysis. Bayesian modeling and data analysis with *Stan* and *JAGS*. Incorporating version control via *Git/GitHub*.

Cognitive Modeling: Applying, testing, and developing diffusion models and its variations: Levy Flight Model, one-boundary diffusion with increasing drift, EZ-Diffusion.

Bayesian Statistics: Hierarchical and non-hierarchical Bayesian modeling, MCMC-based inference (JAGS, Stan), simulation-based inference (BayesFlow), model comparison, and posterior predictive checks.+

English: Professional proficiency. **German:** Beginner, finished A2-level course.

Publications

Hato, T, Schumacher L, Radev S, Voss A. 2024. Lévy Versus Wiener: Assessing the Effects of Model Misspecification on Diffusion Model Parameters. *Comput Brain Behav* 8, 588–606 (2025). <https://doi.org/10.1007/s42113-025-00248-6>

Hato, T, Kilic, A. 2022. Testing Encoding and Retrieval Dynamics of Pairs Using Probed Recall Task (Master thesis; <https://open.metu.edu.tr/handle/11511/98780>)

UNDER REVIEW Pfandt J, Bartos F, Goodman H, ... **Hato T**, Wagenmakers E.J. A Methodological Metamorphosis: The Rapid Rise of Bayesian Inference and Open Science Practices in Psychology

Ongoing Projects

Modeling the EPIC dataset: we modified and applied the One-boundary DDM (Ratcliff & van Dongen, 2011) to the EPIC-Norfolk dataset (N = 8623; ages 48-92) to examine the relationship between response time (RT) measures, demographic factors and future dementia risk over 10 year follow-up (9% diagnosed). Analyses were conducted in three stages: (1) cognitive modeling using simulation-based inference, (2) survival analyses of time-to-dementia outcomes, and (3) supervised machine learning to predict the dementia diagnoses using cognitive model parameters. Manuscript is under preparation.

Time varying parameters with EZ-Diffusion: Aim of this project is to apply EZ-Diffusion to investigate parameter changes over time in an experiment by incorporating random walk priors in a JAGS model.

Modeling large-scale cognitive data from Lumosity: We apply diffusion decision models to multiple reaction-time tasks using simulation-based inference (BayesFlow) to estimate latent decision parameters in large samples. Bayesian structural equation modeling is planned to assess the stability and factor structure of DDM parameters across demographic subgroups.

Teaching Experience

Summer 2025	Seminar , Designed and taught a seminar named Good Practices in Reproducible R Programming for Master's students focusing on data handling and visualization with tidyverse and ggplot libraries.	Heidelberg
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Awards, Fellowships, & Grants

2024	WOMP-PRIME Travel and Networking Award , Society of Mathematical Psychology	€500
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Presentations

INVITED TALKS

Winter 2024, *Lévy versus Wiener: Assessing the Effects of Model Misspecification on Diffusion Model Parameters*. Talk at ProSem, Syracuse University, Syracuse, New York, USA.

CONFERENCE CONTRIBUTIONS (TALKS)

Hato, T. (2025, March 9-12). *EPIC Data, EPIC Models: Unraveling Cognitive Decline with Large-Scale Diffusion Modeling*. Invited talk: 58th Annual Meeting of the Society for Mathematical Psychology (MathPsych) and the 23rd International Conference on Cognitive Modeling (ICCM) at The Ohio State University in Columbus, Ohio, USA.

Hato, T. (2024, July 18-22). *Bias Against Levy Flight: What Happens When We Misspecify Levy Flight as Diffusion Model?*. Invited talk: Joint 57th Meeting of the Society for Mathematical Psychology, 22st International Conference on Cognitive Modeling (MathPsych / ICCM). Tilburg, the Netherlands.

Hato, T. (2024, March 17–20). Investigating The Possible Bias Against Lévy Flight Model in Diffusion Model Paradigm [Talk]. 66th Conference of Experimental Psychologists (Tagung experimentell arbeitender Psycholog:innen, TeaP), Regensburg, Germany.

POSTER PRESENTATIONS

- Hato, T.** 2025. Applying the Diffusion Model to Visual Processing for Understanding Dementia Risk: Insights from the EPIC-Norfolk Cohort. Poster presented at 67th Conference of Experimental Psychologists (Tagung experimentell arbeitender Psycholog:innen, TeaP), Frankfurt (Main), Germany.
- Hato, T.** 2024. Lévy versus Wiener: Assessing the effects of model misspecification on diffusion model parameters. Poster presented at the 65th Annual Meeting for Psychonomic Society, New York, The USA.
- Hato, T.** 2024. Lévy versus Wiener: Assessing the effects of model misspecification on diffusion model parameters. European Mathematical Psychology Group (EMPG), Tübingen, Germany.
- Hato, T.** 2023. Contiguity Effect is Asymmetric Across Pairs, How About Within? Poster presented at the Joint 56th Meeting of the Society for Mathematical Psychology, 21st International Conference on Cognitive Modeling, and 52nd Meeting of the European Mathematical Psychology Group (MathPsych / ICCM / EMPG). Amsterdam, the Netherlands.

Professional Development

Model-Based Neuroscience and Cognition Summer School by Andrew Heathcote and Dora Matzke. Attended this summer school on Bayesian evidence accumulation models especially Linear Ballistic Accumulation (LBA) model.

Bayesian Cognitive Modeling, workshop by E.J. Wagenmakers and Michael Lee. Attended this workshop on Bayesian inference, learning theoretical concepts and practical skills for applying JAGS software to various statistical models. Gained hands-on experience using R with JAGS to implement Bayesian models for cognitive science applications.

PROFESSIONAL MEMBERSHIPS

- Trainee Member, Society of Mathematical Psychology
- Women of Mathematical Psychology
- Graduate Student Member, Psychonomic Society
- DFG Research Training Group: Statistical Modeling in Psychology (SMiP)

PEER REVIEW

- Journal of Experimental Psychology

Academic References

- Prof. Dr. Andreas Voss, Psychological Methods Department, Heidelberg University, andreas.voss@psychologie.uni-heidelberg.de
- Assistant Prof. Dr. Stefan T. Radev, Cognitive Science Department, Rensselaer Polytechnic Institute, radevs@rpi.edu